Report on Cooperative Optimization Project

March 2026

1 Part I

1.1 Decentralized gradient descent

1.1.1 Justification of the step size

Let denote the agent $A_1, A_2, \ldots, A_5$ The function f_i for each agent is:

$$f_i(\alpha) = \frac{\sigma^2}{5}\frac{1}{2}\alpha^T K_{mm}\alpha + \frac{1}{2}\sum_{i\in A_i}(y_i - K_{(i)m}\alpha)^2 + \frac{\nu}{10}\|\alpha\|^2 \tag{1}$$

One has:

$$\nabla_\alpha f_i(\alpha) = \frac{\sigma^2}{5}K_{mm}\alpha - \sum_{i\in A_i} K_{(i)m}^T(y_i - K_{(i)m}\alpha) + \frac{\nu}{5}\alpha \tag{2}$$

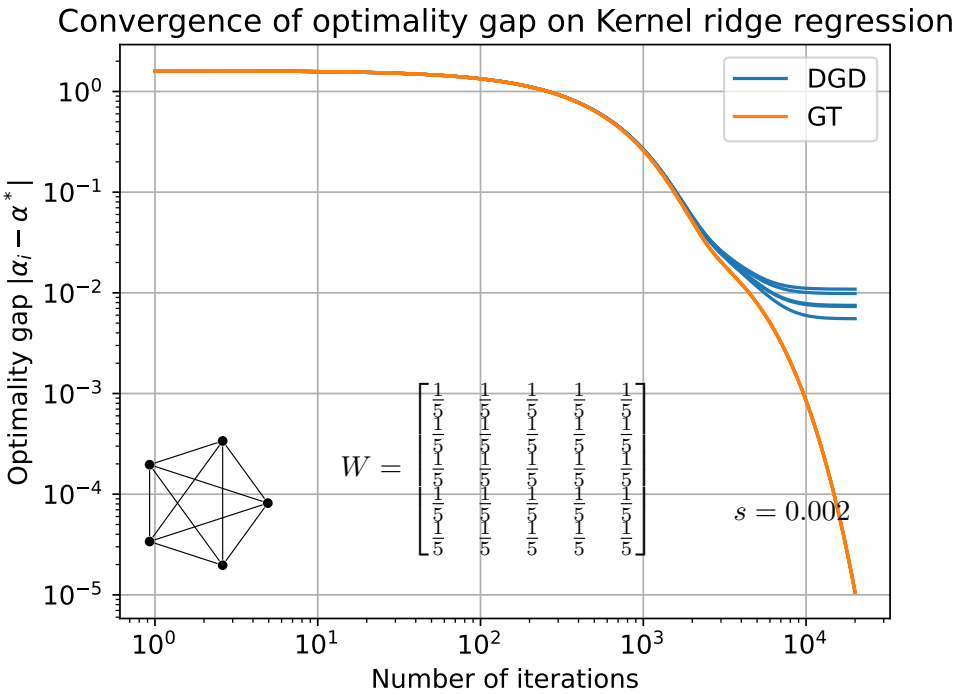
Then:

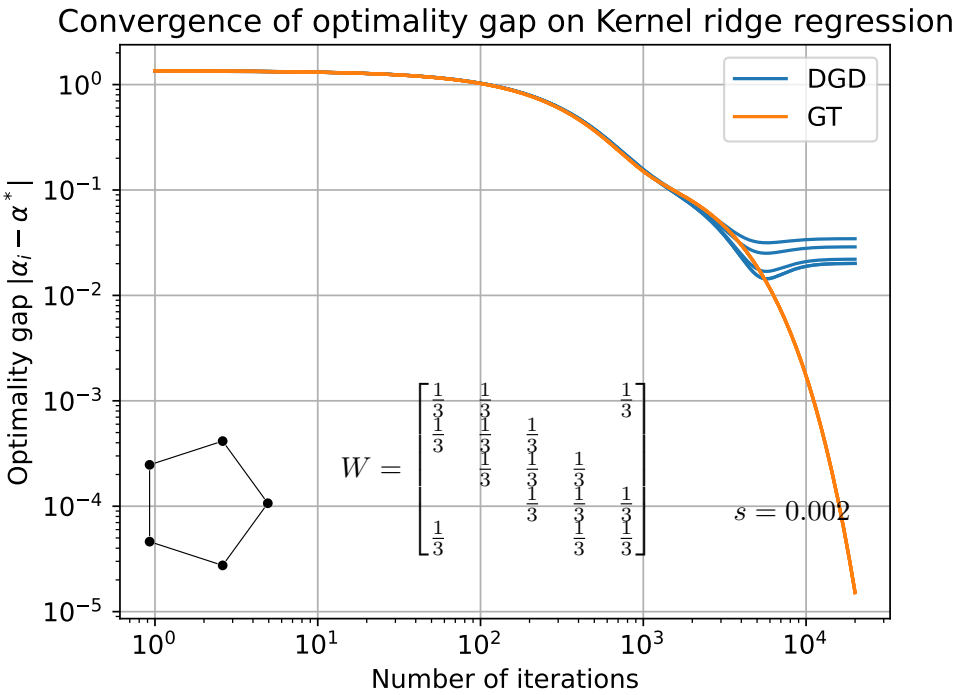
$$\nabla_\alpha^2 f_i(\alpha) = \frac{\sigma^2}{5}K_{mm} + \sum_{i\in A_i} K_{(i)m}^T K_{(i)m} + \frac{\nu}{5}I_m \tag{3}$$

So, let $\lambda_{i,\max} = \arg\max_{\lambda\in\mathrm{Sp}\left(\nabla_\alpha^2 f_i(\alpha)\right)} |\lambda|$ for each agent i. Let $L = \max_i |\lambda_{i,\max}|$. Then, we need to choose a step size which fulfilled is $\leq O\left(\frac{1}{L}\right)$.

1.1.2 Plot convergence depending on graph structure

We observe that the more is connected the graph, the faster is convergence and the more there is consensus.


Convergence of optimality gap on Kernel ridge regression
DGD
GT
Optimality gap $|\alpha_i - \alpha^*|$
Number of iterations
$W = \begin{bmatrix} \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & \frac{1}{5} \\ \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & \frac{1}{5} \\ \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & \frac{1}{5} \\ \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & \frac{1}{5} \\ \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & \frac{1}{5} \end{bmatrix}$
$s = 0.002$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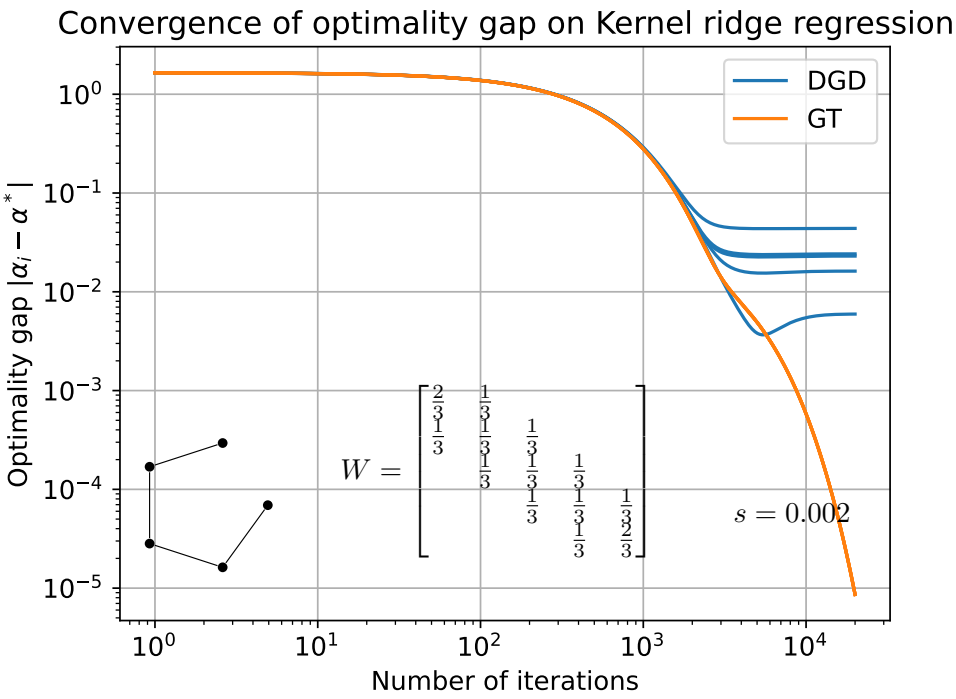
Number of iterations
$W = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & & & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & & \\ & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & \\ & & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & & & \frac{1}{3} & \frac{1}{3} \end{bmatrix}$
$s = 0.002$


Convergence of optimality gap on Kernel ridge regression
DGD
GT
Optimality gap $|\alpha_i - \alpha^*|$
Number of iterations
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^1
10^2
10^3
10^4
$W = \begin{bmatrix} \frac{2}{3} & \frac{1}{3} & & & \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & & \\ & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & \\ & & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ & & & \frac{1}{3} & \frac{2}{3} \end{bmatrix}$
$s = 0.002$